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Computing Project

PIG: GPU Redesign of a Plenoptic Imaging Pipeline and Extension to Multi-View Reconstruction

Niccolò Benedetto
Enrollment number 000626872

Prof. Daniele Bonatto
PhD Ferreira Ribeiro Brenno

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Abstract

This report presents the redesign of an existing CPU plenoptic image generation (PIG) pipeline toward a GPU implementation using CUDA. This work focuses on the acceleration about the most computationally expensive stages of the existing pipeline,and in the same time aims to to preser the rendering correctness. In addition, the new GPU pipeline is extended to support multi-view datasets (RGB + depth) by means GPU-based point cloud registration and merging. Experimental results demnstrate significant runtime improvements and validate the geometric consistency and efficency of the proposed multi-view pipeline regarding the single-view disocclusion problems.

1 Introduction

1.1 Background Context and Motivation

Integral imaging is a promising technique for glasses-free three-dimensional (3D) displays, as it reconstructs the light field, stored as a grid of micro-images, or a plenoptic image, using a microlens array (MLA) placed in front of a display screen. Unlike conventional imaging pipelines, plenoptic rendering requires the generation and manipulation of dense point cloud representations, followed by hard computationally rendering and post-processing stages.

The existing PIG software already provides a complete CPU-based pipeline capable of generating plenoptic images starting from RGB + Depth datasets. However, several stages of the pipeline takes significant computational time, especially in large point clouds. In particular, the point cloud generation and post-processing stages represent the biggest performance bottlenecks due to thei large amount of data computed.

That is the main motive that takes into account the migration toward a GPU-based pipeline. With its parallel computational skills a GPU proofs itself particulary suitable for point cloud oprations and geometric transformations. Hence the redesigning of the original CPU-based implementation into the GPU one represent the natural step my supervisor suggested in order to achieve a better runtime performance.

In addition, if we consider a plenoptic reconstruction approaches that rely on multi-view acquisition setups to reduce disocclusions and improve geometric completeness. By combining RGB + Depth observations acquired from different viewpoints, it becomes possible to reconstruct scene regions that would otherwise remain occluded in a single-view configuration.

1.2 Project Objectives

The first objective of this project is to redesign the most time computationally expensive stages of the original pipeline into a GPU execution while preserving rendering correctness and visual consistency with the existing CPU ground truth implementation and outcomes.

The second objective is to design and implement a new GPU-based multi-view reconstruction pipeline capable of:

- loading and reding correctly synchronized multi-view RGB + Depth datasets;

- generating one point cloud per camera view;
- registering point clouds into a shared world coordinate system bye means rigid body transformations;
- merging the resulting point clouds into a unique geometric point cloud suitable for plenoptic rendering.

A final step consists in validating both the computational performance improvements and the geometric correctness of the proposed multi-view approach through a time and metric analysis.